2015001,2015002,2015003,
2015004,2015005,2015006,
2015007

IoT Based Baby Monitoring System & Smart Cradle
Baby Crying Detector

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Introduction & Motivation

The objectives of this project are as follows-

- To design a smart automated cradle by detecting baby's cry.
- To monitor baby's body temperature, diaper wetness.
- To apply Internet of Thing (IoT) for this monitor system by using Telegram application to notify parents.

The project is to prepare a IoT based smart baby cradle[1] for the babies and their parents who face problem to take care of their children in their day to day life because of their jobs.

Project Description

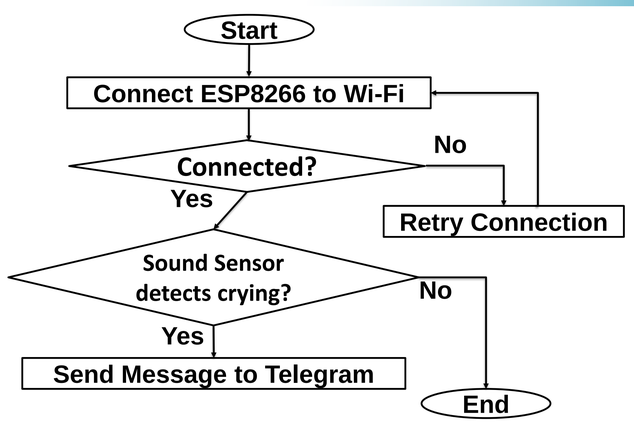


Fig 1. Complete Hardware Setup

Materials & Methods

Materials used:

- Arduino Uno
- Battery
- Sound Sensor
- 9g Servo Motor
- NTC Thernistor
- Veroboard
- ESP 8266
- A basket and Stand made by PVC pipe for the prototype of cradle



Results & Analysis

Telegram Bot Mobile Interface:



Status Table:

Parameter	Status Message	Description
Baby Cry Detection	Audio: Crying/ Audio: Quite	Detects when the baby cries and starts the cradle rocking automatically.
Urination Detection	Peeing/Not Peeing	Detects if the baby has urinated using a rain sensor and sends notification through Telegram bot..
Temperature Reading	Temp: 22.50℃	Continuously monitors the surrounding temperature and displays it on the Telegram bot.

Conclusion

- The proposed design successfully detected a baby's simulated cry, triggering the cradle to rock gently using a servo mechanism. It also detected temperature and wetness, sending alerts to the Telegram app, allowing parents to monitor their baby remotely.

The future modifications may include –

- Cost-effective design compared to expensive smart cradles.
- Potential to monitor sick babies by integrating vital signs sensors.
- Compatibility with baby monitoring cameras and enhanced data transmission through IoT, prioritizing safety and comfort.

Acknowledgements

- We would like to thank our supervisor Md. Khorshed Alom sir & Nawaz Talukder Sanglap sir for the encouragement and the freedom to work throughout the whole process.

[1] A. V. Karanjkar and R. Kumawat, "Design of a Smart Baby Cradle Using Blynk and Local Customer Priorities," *SAMRIDDHI - A Journal of Physical Sciences, Engineering and Technology*, vol. 14, no. 2, pp. xx-xx, 2022. Print ISSN: 2229-7111, Online ISSN: 2454-576.